Final Project: A Comparative Analysis of Machine Learning Architectures for Satellite-Based Surface Monitoring

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1 Introduction

In an era of unprecedented global change, modern geoinformatics has transitioned from traditional, static map-making to dynamic, real-time environmental monitoring. This project provides a comprehensive framework for students to apply the fundamentals of Geospatial Artificial Intelligence (GeoAI) to real-world scenarios, leveraging satellite imagery to quantify and analyze Earth surface covers. By integrating open-source tools with advanced machine learning architectures, students move beyond passive observation toward hands-on, structured spatial analysis. This pedagogical approach mirrors professional GeoAI applications, where model development and deployment occur within specialized computational frameworks independently of traditional GIS software. Through this workflow, participants will bridge the gap between raw spectral data and actionable environmental intelligence.

2 Possible Regions of Interest (ROI)

For this project, teams may choose to revisit the study areas analyzed in previous assignments. However, you are also free to select a new Region of Interest (ROI) from the curated list of case studies, such as Mocoa, Hunga Tonga, or the Kakhovka Dam, or propose a custom area that aligns with the project's objectives. You should pick just one image.

3 Supervised Classification and Spatial Prediction

In this phase, we move from basic image analysis to supervised machine learning. You will structure pixel-level information into training datasets, following a standard deployment workflow that operates independently of external GIS software. This approach mirrors real-world GeoAI applications where model development occurs in specialized computational frameworks.

3.1 Data Extraction and Table Construction

Students must transform the multi-spectral Sentinel bands into a non-spatial, tabular format suitable for machine learning training. Using a programming language of your choice,[1] you will develop scripts to read the satellite imagery and extract pixel values located within your training polygons. Each polygon serves as a ground-truth sample representing specific land cover classes within the scene, allowing you to build a structured dataset where each row corresponds to an individual pixel's spectral signature and its associated label.

[1] Common choices include Python (e.g., using `rasterio` and `geopandas`) or R (e.g., using `terra` and `sf`).

- **Format:** Export data as a Tab-Separated Values (TSV) file.
- **Columns:** The table must include Latitude, Longitude, at least eight spectral bands (e.g., B2–B11), and a categorical class label (e.g., "Vegetation", "Water", "Bare soil", etc).

3.2 Model Implementation and Training

Once the training dataset is prepared, students must implement and compare five distinct machine learning architectures. This comparative analysis is designed to evaluate how different mathematical approaches handle the high-dimensional spectral data of your selected scene. You will assess each model's performance using standard validation metrics, including the Confusion Matrix, Kappa Coefficient, and F1-Score.

1. **Decision Trees (DT):** A logic-based model that utilizes a hierarchical tree structure to split data based on spectral thresholds. This approach is particularly useful for identifying which specific bands are most influential in distinguishing land cover classes.
2. **Support Vector Machines (SVM):** A kernel-based algorithm that identifies the optimal hyperplane to maximize the margin between class clusters in a multi-dimensional feature space. It is highly effective for complex classification tasks where classes are not linearly separable.
3. **Artificial Neural Networks (ANN):** A computational model composed of interconnected layers of neurons that learn complex, non-linear relationships between spectral signatures and land cover categories.
4. **K-Nearest Neighbor (KNN):** A non-parametric method that assigns a class membership based on a plurality vote of its neighbors. Pixels are classified according to the most common category among the "k" nearest training samples in the spectral feature space.
5. **Naive Bayes (NB):** A probabilistic classifier based on Bayes' Theorem that assumes strong (naive) independence between the input spectral bands. It calculates the probability of each class given the observed pixel values and selects the most likely outcome.

3.3 Model Comparison and Performance Evaluation

To identify the most robust classification architecture, teams must perform an exhaustive Hyperparameter Optimization (HPO) process. This involves systematically testing and adjusting the internal settings of each algorithm to maximize its predictive power for your specific study area.

Once the models are tuned, you must conduct a rigorous performance assessment by comparing their resulting Confusion Matrices. Students are required to compute and analyze the following evaluation metrics:

- **Overall Accuracy:** To quantify the total percentage of correctly classified pixels.
- **Kappa Coefficient:** To measure the agreement between predicted and reference data beyond what is expected by chance.
- **Precision and Recall:** To assess the model's reliability and completeness for each land-cover category.
- **F1-Score:** To determine the harmonic balance between precision and recall, ensuring the model remains accurate across all classes.

Finally, you must synthesize your findings into a comprehensive summary table. This table should facilitate a direct comparison between the five architectures, clearly highlighting the "best" performance based on the metrics listed above.

3.4 Final Raster Prediction (Inference)

The culmination of the classification pipeline is the generation of a comprehensive thematic land-cover map through full-scene inference. Once the optimal model has been identified and validated, having achieved the highest statistical performance in terms of F1-Score and Kappa Coefficient, it must be deployed to classify the remaining pixels that comprise the rest of the study area.

Unlike the training phase, which utilized a discrete Tab-Separated Values (TSV) dataset derived from specific polygons, the inference phase requires the model to process the original multidimensional raster stack. The model will iterate through each pixel, treating its eight or more spectral bands as input features to predict its most likely class membership. This process transforms raw reflectance data into a new, single-band categorical raster.

The output of this operation must be exported as a georeferenced GeoTIFF. In this final product, each pixel value corresponds to a specific integer code representing its predicted category (e.g., 1 for "Vegetation," 2 for "Water", 3 for "Bare soil" and so on). To ensure the result is of professional quality, students must address the following requirements:

- **Spatial Consistency:** The final raster must retain the original Coordinate Reference System (CRS) and affine transformation parameters. This ensures that the predictions align perfectly with the original satellite imagery and other geospatial layers.
- **Thematic Visualization:** Students must import the resulting GeoTIFF into a GIS environment (such as QGIS) and apply a discrete color symbology. This visualization should clearly distinguish between different land-cover types.
- **Area Quantification:** Beyond visual representation, you are required to generate spatial statistics. Use the predicted raster to calculate the total area (in hectares or square kilometers) for each class.

4 Project Phases and Weekly Deliverables

The project is divided into three milestones, each to be submitted at the end of its respective week:

Phase 1: Problem Definition (Week 1)

Focuses on event selection and data tranformation.

- **Deliverables:** A document defining the chosen event, study area boundaries, satellite imagery metadata, and dataset creation.

Evento elegido

El evento seleccionado para el desarrollo del proyecto corresponde a los incendios forestales ocurridos en enero de 2025 en la ciudad de Los Ángeles, California, Estados Unidos, específicamente en las zonas afectadas por el incendio Palisades y sectores cercanos.

El objetivo del proyecto es aplicar técnicas de Machine Learning y GeoAI para realizar una clasificación supervisada de coberturas terrestres utilizando imágenes satelitales Sentinel-2. A partir de este análisis se busca identificar áreas quemadas, vegetación, zonas urbanas, cuerpos de agua y suelo desnudo dentro del área afectada por los incendios.

Figure 1: Área afectada por los incendios forestales en Los Ángeles, California.

Límites del área

El área de estudio (ROI) se localiza en el oeste de Los Ángeles, California, incluyendo sectores cercanos a Pacific Palisades, Malibu y Topanga.

La delimitación del área fue realizada en QGIS mediante un recorte rectangular aplicado sobre la imagen satelital multibanda, con el objetivo de reducir el tamaño de procesamiento y enfocar el análisis únicamente en las zonas afectadas por los incendios forestales.

Metadatos de la imagen satelital

La imagen utilizada corresponde a una adquisición Sentinel-2 Level-2A del 1 de febrero de 2025 obtenida del programa Copernicus.

Para el análisis se utilizaron ocho bandas espectrales correspondientes a B2, B3, B4, B5, B6, B7, B8 y B11. Las bandas B2, B3, B4 y B8 poseen resolución espacial de 10 metros, mientras que las bandas B5, B6, B7 y B11 originalmente poseen resolución de 20 metros, por lo que fueron remuestreadas a 10 metros mediante interpolación bilineal en QGIS para mantener consistencia espacial entre todas las bandas utilizadas.

Posteriormente, las bandas fueron integradas en un raster multibanda mediante un Virtual Raster (VRT), el cual fue recortado al área de estudio y utilizado para la extracción de información espectral y generación del dataset.

Parámetro	**Valor**
Satélite	Sentinel-2
Fecha de adquisición	2025-02-01
Nivel de procesamiento	Level-2A
Resolución espacial	10 m
Bandas utilizadas	B2, B3, B4, B5, B6, B7, B8, B11
Composición RGB	4-3-2
False Color	8-4-3

Table 1: Metadatos de la imagen satelital utilizada.

Creación del dataset

La construcción del dataset se realizó en QGIS a partir de la imagen multibanda Sentinel-2 y muestras de entrenamiento creadas manualmente sobre el área de estudio.

Inicialmente, se creó una capa vectorial tipo shapefile denominada `training_samples.shp`, utilizada para almacenar polígonos representativos de las diferentes clases de cobertura terrestre presentes en la imagen.

Las clases utilizadas fueron:

- Vegetation
- BurnedArea
- Urban
- Water
- BareSoil

Cada polígono fue dibujado manualmente sobre zonas homogéneas utilizando herramientas de digitalización de QGIS. Posteriormente, mediante la herramienta `Puntos aleatorios en polígonos`, se generaron puntos aleatorios dentro de cada muestra de entrenamiento.

Luego, usando la herramienta `Muestrear valores raster`, se extrajeron automáticamente los valores espectrales correspondientes a las ocho bandas del raster multibanda `sentinel_stack_crop.tif` para cada punto generado.

El resultado fue una nueva capa con información espectral y etiquetas de clase para cada muestra de entrenamiento. Posteriormente, se añadieron las coordenadas X/Y mediante la calculadora de campos y el dataset fue exportado en formato TSV (Tab-Separated Values).

La estructura final del dataset corresponde a:

class	BAND_1	BAND_2	BAND_3	BAND_4	...	BAND_5	BAND_6	BAND_7	BAND_8	X	Y
Water	1030	1005	1009	1023	...	1153	1132	1033	1017	345847.945	3767215.146

Table 2: Estructura del dataset generado en QGIS.

Donde:

- BAND_1 corresponde a la banda B2
- BAND_2 corresponde a la banda B3
- BAND_3 corresponde a la banda B4
- BAND_4 corresponde a la banda B5
- BAND_5 corresponde a la banda B6
- BAND_6 corresponde a la banda B7
- BAND_7 corresponde a la banda B8
- BAND_8 corresponde a la banda B11

Este dataset será utilizado posteriormente para el entrenamiento y evaluación de los modelos de Machine Learning del proyecto.

Phase 2: Pre-processing and Training (Week 2)

Focuses on technical preparation and initial model testing.

- **Deliverables:** Documentation of the exported TSV training file, and initial model evaluation metrics (at least Confusion Matrix and Overall Accuracy).

Preprocesamiento del dataset

El entrenamiento de los modelos fue desarrollado utilizando Python y la librería `scikit-learn` en Visual Studio Code. El dataset generado en QGIS fue exportado en formato CSV y posteriormente cargado para realizar el preprocesamiento y entrenamiento de los modelos de clasificación supervisada.

Inicialmente, se importó el dataset utilizando la librería `pandas`, separando las variables espectrales y las etiquetas de clase.

La columna `class` fue utilizada como variable objetivo para el entrenamiento de los modelos.

Posteriormente, se realizó una normalización de los datos utilizando `StandardScaler`, con el fin de mejorar el rendimiento de modelos sensibles a diferencias de escala como SVM, KNN y ANN.

Luego, el dataset fue dividido en conjuntos de entrenamiento y prueba mediante la función `train_test_split`, utilizando un 70% de las muestras para entrenamiento y un 30% para validación.

Implementación de modelos

Para la clasificación supervisada se implementaron cinco arquitecturas de Machine Learning utilizando la librería `scikit-learn`:

- Decision Tree (DT)
- Support Vector Machine (SVM)
- K-Nearest Neighbor (KNN)
- Naive Bayes (NB)
- Artificial Neural Network (ANN)

Cada modelo fue entrenado utilizando las muestras espectrales obtenidas del dataset. Posteriormente, se realizaron predicciones sobre el conjunto de prueba para evaluar el desempeño de cada arquitectura.

Los modelos fueron implementados con los siguientes parámetros iniciales:

- **Decision Tree:** profundidad máxima de 10 niveles.
- **SVM:** kernel radial basis function (RBF).
- **KNN:** 5 vecinos más cercanos.
- **Naive Bayes:** clasificador Gaussiano.
- **ANN:** red neuronal multicapa con 100 neuronas ocultas y 500 iteraciones máximas.

Evaluación de desempeño

El desempeño de los modelos fue evaluado mediante métricas de clasificación calculadas sobre el conjunto de prueba.

Las métricas utilizadas fueron:

- Accuracy
- Precision
- Recall
- F1-Score
- Kappa Coefficient

Además, para cada modelo se generó una matriz de confusión con el fin de visualizar el comportamiento de clasificación entre las diferentes coberturas terrestres.

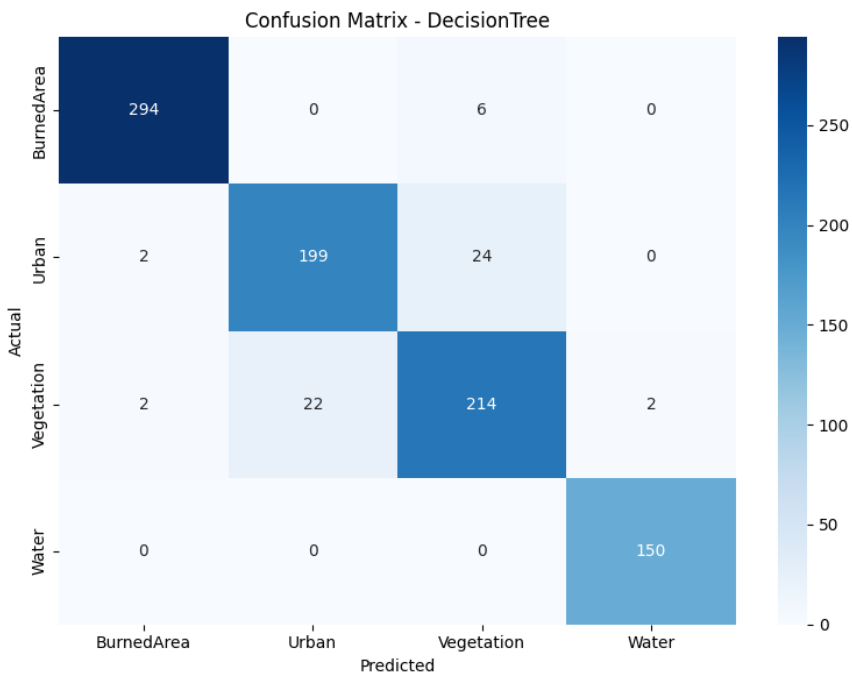

Figure 2: Matriz de confusión del modelo Decision Tree.

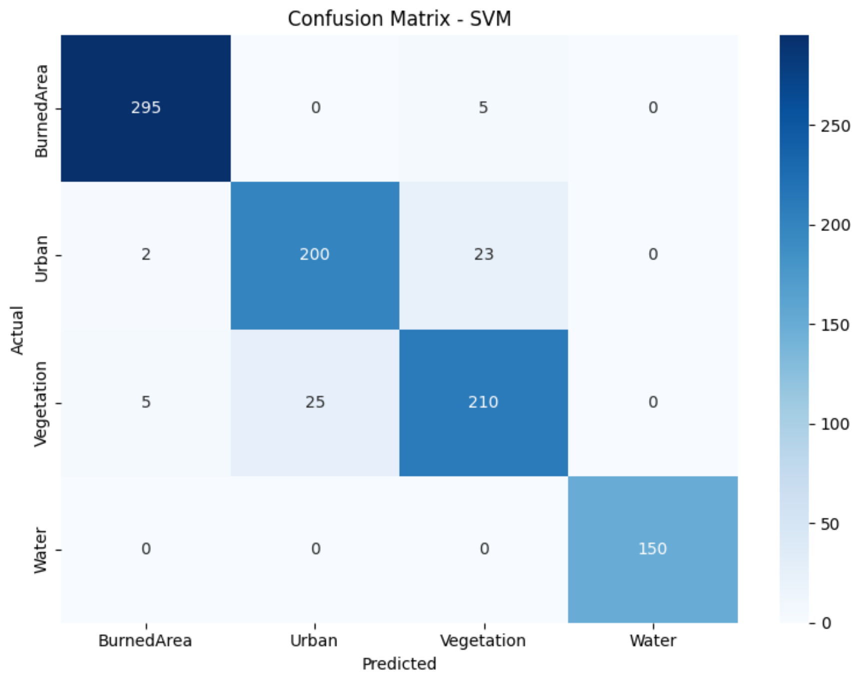

Figure 3: Matriz de confusión del modelo SVM.

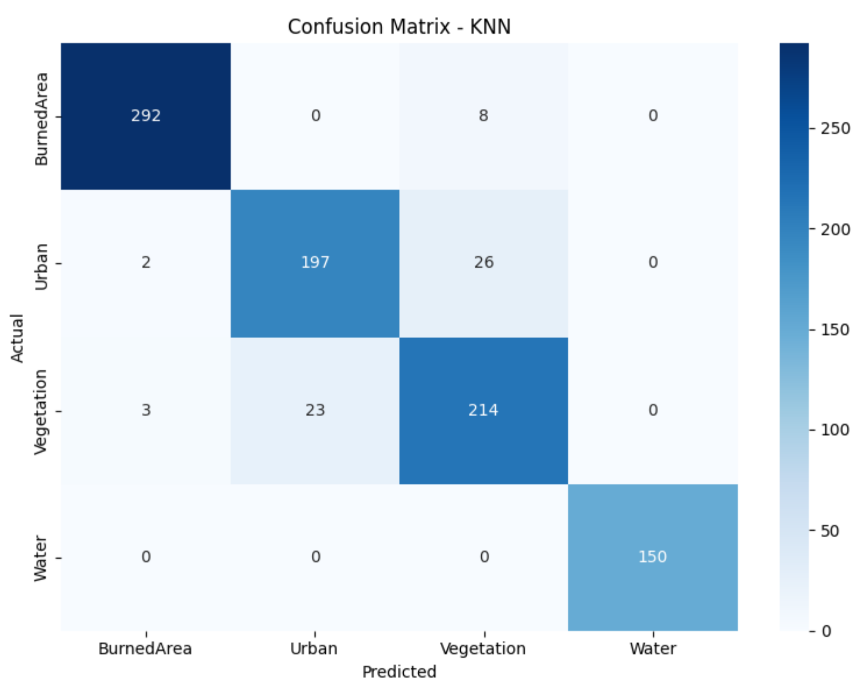

Figure 4: Matriz de confusión del modelo KNN.

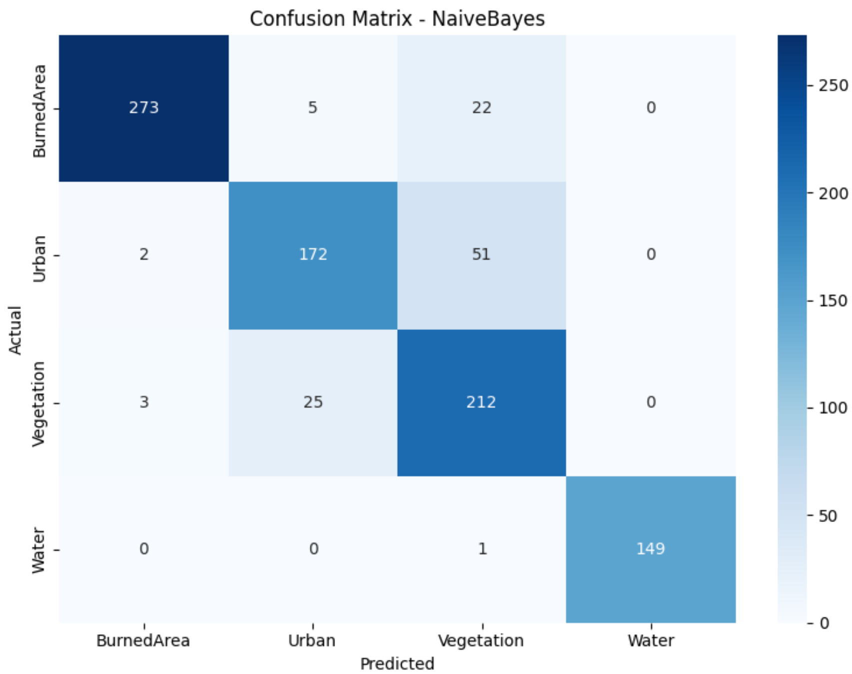

Figure 5: Matriz de confusión del modelo Naive Bayes.

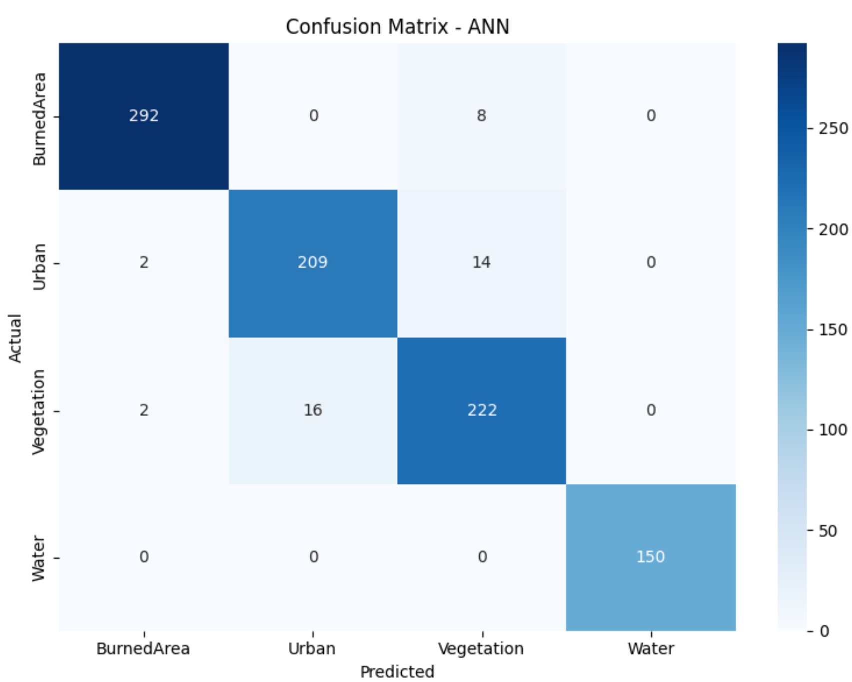

Figure 6: Matriz de confusión del modelo ANN.

Resultados obtenidos

Los resultados obtenidos muestran un alto desempeño general en la clasificación supervisada de coberturas terrestres utilizando información espectral Sentinel-2.

El modelo con mejor desempeño fue Artificial Neural Network (ANN), obteniendo los mayores valores de Accuracy, F1-Score y Kappa Coefficient.

Modelo	**Accuracy**	**F1-Score**	**Kappa**
Decision Tree	0.936	0.936	0.913
SVM	0.934	0.934	0.910
ANN	0.954	0.954	0.937
KNN	0.932	0.932	0.907
Naive Bayes	0.880	0.882	0.838

Table 3: Comparación de desempeño de los modelos de Machine Learning.

Los resultados evidencian que los modelos basados en aprendizaje no lineal, especialmente ANN, presentan una mayor capacidad para diferenciar coberturas complejas como áreas urbanas, vegetación y superficies quemadas utilizando información espectral multibanda.

Por otro lado, Naive Bayes presentó el menor desempeño debido a la suposición de independencia entre variables espectrales, condición que no se cumple completamente en imágenes multiespectrales Sentinel-2.

Phase 3: Synthesis and Prediction (Week 3)

Final analysis and reporting.

- **Deliverables:** The predicted thematic raster (GeoTIFF), the complete technical report, and the source code (ZIP).

Final Defense and Evaluation

The project results will be presented in a formal oral defense during the last session of the course. To ensure collective technical mastery, two team members will be randomly selected to lead the presentation at the start of the session, so everyone should be ready to step into the spotlight. Teams must prepare professional slides and supporting documentation for a presentation not exceeding 20 minutes. The evaluation will follow a collaborative framework, incorporating both auto-evaluation (self-assessment) and co-evaluation (peer review) to assess technical accuracy, methodology, and communication clarity.

5 Conclusion

This project represents a complete synthesis of the remote sensing pipeline, from initial data acquisition to high-level machine learning inference. By navigating the complexities of multi-spectral data extraction , Hyperparameter Optimization (HPO) , and full-scene prediction, students transform raw reflectance values into a meaningful, single-band categorical raster. Beyond the technical mastery of scripts and algorithms , the true value of this work lies in the ability to document our changing planet using ground-truth validation and scientific integrity. As students finalize their thematic visualizations and quantitative area assessments, they transition from being observers of environmental flux to active participants in the modern geospatial landscape.